

Given that the boiling point of Compound 1 is 18 °C, the boiling point of Compound 2 would most likely be:

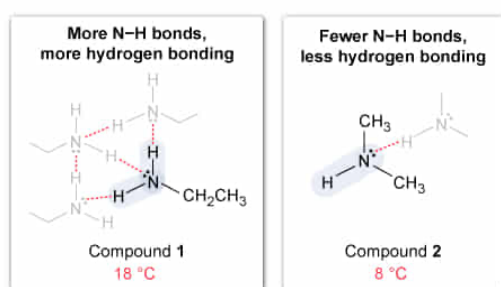
- ✓ ☒ A. 8 °C. [61%]
☐ B. 18 °C. [23%]
☐ C. 89 °C. [14%]
☐ D. 100 °C.

Correct

61% answered correctly

Explanation:

Boiling point is influenced by hydrogen bonding



Boiling point decreases as the hydrogen bonding decreases

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Intermolecular forces cause attractions between molecules. Structures with a **lesser** extent of **intermolecular forces** have **lower boiling points** because less energy is required to overcome these attractions between molecules during the **phase change** from a liquid state to a gas state.

Hydrogen atoms covalently bonded with nitrogen, oxygen, or fluorine atoms form **polar bonds** with partial charges that produce **dipoles**. Attractions between partial charges of opposite signs generate a significant intermolecular force known as **hydrogen bonding**. This term does not refer to the covalent chemical bond between hydrogen and another atom but instead refers to the attractions between the hydrogen bond dipoles.

When comparing similar molecular structures, molecules with fewer hydrogen-bonding groups will have fewer hydrogen bonding interactions. For example, Compound 1 has two N–H bonds capable of hydrogen bonding, whereas Compound 2 has only one N–H bond. Having fewer N–H bonds decreases the extent of hydrogen bonding and lowers the energy required to overcome the intermolecular attractions. As a result, the boiling point of Compound 2 is lower than that of Compound 1.

Therefore, out of the choices in this question, the most likely boiling point of Compound 2 is 8 °C.

(Choice B) Although Compounds **1** and **2** are constitutional isomers with the same composition and molar mass, because these compounds have different extents of hydrogen bonding, their boiling points cannot be the same.

(Choices C and D) Because Compound **2** has fewer N–H bonds (resulting in less hydrogen bonding), its boiling point is *lower* than that of Compound **1**.

Educational objective:

Structures with a lesser extent of intermolecular forces have lower boiling points because less energy is required to overcome the attractions between molecules. When comparing similar structures, the boiling point tends to decrease as the number of hydrogen-bonding groups decreases due to fewer hydrogen bonding interactions.

Time spent:0 sec

Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws